Roya Khalili Amirabadi

Applied Mathematics | Optimization | Reinforcement Learning

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Google Scholar | LinkedIn

Research Profile

Applied mathematician with a Ph.D. in Operations Research and experience in academic research, university teaching, and industrial R&D. Research focuses on reinforcement learning for constrained nonlinear optimal control, uncertain dynamical systems, and optimization. Publications address safe actor–critic control, experience replay, LSTM-based modeling, and fuzzy sliding mode control. Current industrial work explores data-driven predictive maintenance and physics-informed neural networks for steel processes. Uses Python and MATLAB for model development, numerical simulation, and analysis.

Education

Ph.D., Applied Mathematics (Operations Research) Sep 2020 – Aug 2025

Ferdowsi University of Mashhad, Iran

GPA: 19.08/20 | Dissertation grade: 20/20

Solving a Class of Optimal Control Problems of Continuous-Time Systems Using Deep Learning Techniques

Supervisor: Dr. Omid Soleimanifard; Co-advisor: Dr. Mohsen Jalaeian Farimani

M.Sc., Applied Mathematics (Operations Research) Sep 2011 – Feb 2014

Payame Noor University of Mashhad, Iran

GPA: 17.25/20 | Thesis grade: 19.25/20

Sliding Mode Control of Outbreaks of Emerging Infectious Disease

Supervisor: Dr. Aghileh Heydari; Co-advisor: Dr. Mohammad Reza Zarrabi

B.Sc., Applied Mathematics Feb 2004 – Jul 2008

Payame Noor University, Torbat-e Heydarieh, Iran

Industrial Research Experience

R&D Team Member Sep 2025 – Present

Knowledge-Based Automation and Project Development Industrial Company of Paz

- Develop data-driven and learning-based models for industrial systems, supported by Python-based simulation and analysis of operational datasets.
- Investigate reinforcement learning approaches to predictive maintenance in the steel industry.
- Explore physics-informed neural networks for unmodeled dynamics in steel heating and melting processes.
- Relate modeling and machine learning methods to engineering and operational objectives.

Research Expertise

- **Learning-based control:** safe reinforcement learning, actor–critic methods, experience replay, and LSTM-assisted control of nonlinear systems with uncertain dynamics.
- **Optimization and robust control:** constrained optimal control, hybrid metaheuristics, non-stationary optimization, and fuzzy sliding mode control.
- **Applications and synthesis:** epidemic models, industrial process modeling, and systematic reviews of experience replay and reinforcement learning–MPC integration.

Peer-Reviewed Journal Articles

1. **Khalili Amirabadi, R.**, Jalaeian-Farimani, M., & Solaymani-Fard, O. (2026). Experience replay in reinforcement learning: A systematic review. *Neurocomputing*, 700, 134403. DOI.
2. **Khalili Amirabadi, R.**, Solaymani-Fard, O., Mansoori, A., & Jalaeian-Farimani, M. (2026). Improved control of uncertain chaotic systems using hybrid fuzzy sliding mode techniques: Chattering elimination and robust performance. *Fuzzy Sets and Systems*, 109944. Publisher record.
3. **Khalili Amirabadi, R.**, Jalaeian-Farimani, M., & Solaymani-Fard, O. (2026). Self-organizing dual-buffer adaptive clustering experience replay (SODACER) for safe reinforcement learning in optimal control. *Scientific Reports*, 16, 14960. DOI.
4. **Khalili Amirabadi, R.**, Jalaeian-Farimani, M., & Solaymani-Fard, O. (2025). LSTM-empowered reinforcement learning in bi-level optimal control for nonlinear systems with uncertain dynamics. *ISA Transactions*. Publisher record.
5. **Khalili Amirabadi, R.**, Solaymani-Fard, O., & Jalaeian-Farimani, M. (2025). Towards optimal control of HPV model using safe reinforcement learning with actor–critic neural networks. *Expert Systems with Applications*, 264, 125783.
6. Jalaeian-Farimani, M., **Khalili Amirabadi, R.**, Esmaeili Ranjbar, M., & Samadzadeh, S. (2025). Event-triggered dynamic seed invasive weed optimization (ET-DSIWO): A nature-inspired approach for non-stationary optimization. *Nonlinear Dynamics*, 113(20), 27611–27636. Institutional record.
7. **Khalili Amirabadi, R.** & Solaymani-Fard, O. (2024). Combining hybrid metaheuristic algorithms and reinforcement learning to improve the optimal control of nonlinear continuous-time systems with input constraints. *Computers and Electrical Engineering*, 116. Publisher record.
8. **Khalili Amirabadi, R.**, Solaymani-Fard, O., & Mansoori, A. (2021). A novel fuzzy sliding mode control approach for chaotic systems. *Iranian Journal of Fuzzy Systems*, 18(6), 133–150.
9. **Khalili Amirabadi, R.**, Heydari, A., & Zarrabi, M. R. (2016). Analysis and control of SEIR epidemic model via sliding mode control. *Advanced Modeling and Optimization*, 18(1), 153–162. Full text.

Preprint

Jalaeian-Farimani, M., **Khalili Amirabadi, R.**, Nikkhouy, D., Abdolbaghi, M., Rastegarmoghaddam, M., Samadzadeh, S., & Ghane, M. (2026). A systematic review and taxonomy of reinforcement learning-model predictive control integration for linear systems. arXiv:2604.21030.

Earlier versions of the SODACER journal article are available as arXiv:2601.06540 and SSRN 5191427; they are not listed as separate publications.

Book Chapters – In Press

Forthcoming contributions (2026):

1. Jalaeian Farimani, M., **Khalili Amirabadi, R.**, Pakdaman, M., & Salahshour, S. Intelligent multi-agent control and forecasting for IoT-driven smart farming: Integrating model predictive control and reinforcement learning.
2. Pakdaman, M., Jalaeian Farimani, M., & **Khalili Amirabadi, R.** Climate-smart and smart agriculture: A review.
3. Jalaeian Farimani, M., **Khalili Amirabadi, R.**, & Pakdaman, M. A survey of industrial predictive control environments for autonomous agricultural robotics and smart greenhouse management.

All in A. Ahmadian (Ed.), *AIoT for Smart Farming: Intelligence, Innovation, and Sustainability.* IET/Wiley.

Teaching Experience

University Instructor, Iran 2014 – 2020

Taught undergraduate Fundamental Mathematics and Artificial Intelligence; prepared teaching materials and mentored students.

University Teaching in Mathematics, Iran 2008 – 2011

Teaching Assistant, Ferdowsi University of Mashhad

Courses: Advanced Mathematics; Differential Equations; Numerical Analysis; Reinforcement Learning; Nonlinear Optimization.

Technical Skills

- **Programming and computation:** Python, MATLAB, Mathematica; scientific computing, numerical simulation, data analysis, and visualization.
- **Mathematical methods:** nonlinear optimization, optimal control, fuzzy systems, sliding mode control, and dynamical systems modeling.
- **Machine learning:** deep reinforcement learning, actor–critic networks, LSTM models, experience replay, and physics-informed neural networks.
- **Research communication:** scientific writing, systematic reviews, and technical presentations.

Academic Distinctions

- Ranked first in academic performance among Ph.D. students, Ferdowsi University of Mashhad.
- Ranked first in the M.Sc. program, Payame Noor University of Mashhad.
- Ranked third nationwide among Payame Noor University M.Sc. students.

Selected Training and Conference Participation

- **MATLAB training (basic and advanced):** Arg Institute, 25 hours, February 2016.
- **Poster presentation:** International Conference on Research in Science and Technology, Kuala Lumpur, 2015.
- **Poster presentation:** National Conference on Mathematics of Payame Noor University, Isfahan, 2014.

Languages

Persian: Native | English: C1 | French: Basic reading and comprehension

Academic References

Dr. Omid Soleimanifard
Associate Professor, Department of Mathematical Sciences
Ferdowsi University of Mashhad, Iran
Soleimani@um.ac.ir

Ph.D. supervisor | Academic profile

Dr. Mohsen Jalaeian Farimani
Assistant Professor, Department of Electronics, Information and Bioengineering (DEIB)
Politecnico di Milano, Italy
mohsen.jalaeian@polimi.it

Ph.D. co-advisor | Academic profile

Dr. Sohrab Effati
Professor
Ferdowsi University of Mashhad, Iran
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Academic profile

Dr. Amin Mansoori
Assistant Professor
Ferdowsi University of Mashhad, Iran
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Academic profile

Dr. Aghileh Heydari
Professor, Department of Mathematics
Payame Noor University of Mashhad, Iran
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M.Sc. supervisor | Academic profile

Dr. Mohammad Reza Zarrabi
Assistant Professor, Department of Applied Mathematics,
School of Mathematics and Computer Sciences
Damghan University, Iran
mzarrabi@du.ac.ir

M.Sc. co-advisor | Research profile